Bellabeat Wellness

Smart Device Data Analysis and Marketing Insights

Tools: SQL (BigQuery) • R / RStudio • Tableau

34	March to April 2016	396	Kaggle / FitBit
Users Tracked	Date Range	Total Records	Source

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Dataset: FitBit Fitness Tracker Data, kaggle.com/datasets/arashnic/fitbit

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Phase 1 : ASK

BUSINESS TASK & STAKEHOLDERS

1.1 Company Background

Bellabeat is a women-focused health technology company founded in 2013 by Urska Srsen and Sando Mur. The company designs smart wellness products that collect data on activity, sleep, stress, and reproductive health, empowering women with meaningful and personalized insights about their bodies and daily habits.

Since its founding, Bellabeat has grown into a purpose-driven brand with a global presence, selling products through its own website, online retailers, and select physical stores. The company invests heavily in digital marketing across platforms including Google Search, Instagram, Facebook, Twitter, and YouTube, while also maintaining a presence in traditional media.

Bellabeat's core product ecosystem includes:

- **Bellabeat App** tracks activity, sleep, stress, menstrual cycle, and mindfulness, connecting to all Bellabeat hardware as the central hub
- **Leaf** is a versatile wellness tracker worn as a bracelet, necklace, or clip
- **Time** is a wellness watch that blends classic design with smart activity and sleep tracking
- **Spring** is a smart water bottle that monitors daily hydration levels
- **Bellabeat Membership** is a subscription service offering personalized wellness guidance based on individual user data

With a strong brand identity and a loyal customer base, Bellabeat is well positioned to expand its share of the global smart device market, and data-driven marketing is central to that growth strategy.

1.2 Business Task

Analyze smart device usage data from FitBit users to identify trends in how consumers use non-Bellabeat smart devices. Apply these insights to one Bellabeat product to guide and improve Bellabeat's marketing strategy.

Recommendations focus on the Bellabeat App, which aggregates data across all Bellabeat hardware products and has the widest marketing reach.

1.3 Key Questions

This analysis is guided by three core business questions, each building on the last, from observation to interpretation to action.

Question	Business Question
Q1	What are the most common trends in smart device usage among FitBit users?
Q2	How could these usage trends apply to Bellabeat customers?
Q3	How could these insights influence and improve Bellabeat's marketing strategy?

Q1 is answered through data. Q2 bridges the FitBit findings to the Bellabeat context. Q3 translates everything into concrete and actionable marketing recommendations, which is the ultimate deliverable of this case study.

1.4 Stakeholders

Stakeholder	Role
Urska Srsen	Co-founder & Chief Creative Officer, primary decision maker
Sando Mur	Co-founder & Mathematician, part of executive team
Marketing Analytics Team	Responsible for collecting, analyzing, and presenting data insights

Phase 2 : PREPARE

DATASET ASSESSMENT & CREDIBILITY

2.1 Dataset Overview

The FitBit Fitness Tracker Data (Kaggle, CC0 Public Domain) was collected from 30–35 consenting FitBit users between March 12 and April 12, 2016, via a distributed Amazon Mechanical Turk survey. The dataset contains 18 CSV files covering minute-level physical activity, heart rate, sleep, steps, and weight. Four files were selected for this analysis.

Dataset source: kaggle.com/datasets/arashnic/fitbit

2.2 Dataset Contents

File	What It Contains
dailyActivity_merged.csv	Daily steps, distance, calories, active minutes
dailyCalories_merged.csv	Daily calorie burn
dailyIntensities_merged.csv	Time spent in each intensity zone per day
dailySteps_merged.csv	Daily step count
heartrate_seconds_merged.csv	Heart rate every second
hourlyCalories_merged.csv	Calories burned per hour
hourlyIntensities_merged.csv	Activity intensity per hour
hourlySteps_merged.csv	Steps taken per hour
minuteCaloriesNarrow_merged.csv	Calories per minute (narrow format)
minuteCaloriesWide_merged.csv	Calories per minute (wide format)
minuteIntensitiesNarrow_merged.csv	Intensity per minute (narrow format)
minuteIntensitiesWide_merged.csv	Intensity per minute (wide format)
minuteMETsNarrow_merged.csv	Metabolic equivalent per minute
minuteSleep_merged.csv	Sleep data per minute
minuteStepsNarrow_merged.csv	Steps per minute (narrow format)
minuteStepsWide_merged.csv	Steps per minute (wide format)
sleepDay_merged.csv	Daily sleep records
weightLogInfo_merged.csv	Weight, BMI

Four files were selected for this analysis based on their direct relevance to the business questions. The dailyActivity_merged.csv file served as the primary analysis file containing daily steps, distance, calories, and active minutes. The sleepDay_merged.csv file was used for sleep pattern analysis covering total sleep time and time in bed. The hourlySteps_merged.csv file enabled hourly activity trend analysis to identify peak usage windows throughout the day. The weightLogInfo_merged.csv file was included to assess weight tracking behavior and feature adoption patterns.

2.3 ROCCC Credibility Assessment

ROCCC Criteria	Rating	Evaluation
Reliable	Low	Only 33 to 35 users. Insufficient for broad generalization. Noted as a key limitation.
Original	No	Third-party data via Amazon Mechanical Turk, not collected directly by Bellabeat.
Comprehensive	Moderate	Covers activity, sleep, calories, steps, heart rate, weight. Missing demographics and gender.
Current	No	Data from 2016. Consumer behavior and device capabilities have changed significantly.
Cited	Yes	Publicly available on Kaggle with clear source attribution.

Important note for this analysis:

Despite these limitations, the dataset still reveals meaningful patterns in human activity and sleep behavior that are relevant to Bellabeat's marketing strategy. All recommendations in this report are made with these limitations clearly in mind and should be validated with more current and larger datasets before being implemented as business decisions.

Phase 3 : PROCESS

DATA CLEANING & DOCUMENTATION

3.1 Data Exploration

Before cleaning any data, I ran initial exploration queries in BigQuery to understand the structure, size, and quality of the dataset.

Query 1 : Row Counts and Date Range

Purpose: Confirm dataset size and collection timeframe.

total_rows	unique_users	earliest_date	latest_date
457	35	2016-03-12	2016-04-12

Finding: The daily activity table contains 457 records from 35 unique users collected over one month. This confirms the expected collection window stated in the source documentation.

Query 2 : Unique Users Per Table

Purpose: Understand how many users participated in each feature.

daily_users	sleep_users	hourly_users	weight_users
35	24	34	11

Finding: User participation drops sharply from automatic tracking to manual logging. Almost every user tracked steps passively but only 68% recorded sleep and just 31% logged weight. Users engage with features that require no effort but disengage from features requiring deliberate action. This behavioral gap is a key insight for Bellabeat's marketing strategy.

Query 3 : Zero Activity Days

Purpose: Identify days where the device was likely not worn.

zero_activity_days
61

Finding: 61 out of 457 records show zero steps or zero calories burned. These rows represent days where the device was not worn or not synced properly. They do not reflect real user activity and will be removed during the cleaning stage to ensure accurate analysis.

Query 4 : Duplicate Check in Sleep Data

Purpose: Find repeated sleep records for the same user on the same date.

Id	SleepDay	duplicate_count
4388161847	5/5/2016 12:00:00 AM	2
4702921684	5/7/2016 12:00:00 AM	2
8378563200	4/25/2016 12:00:00 AM	2

Result: 3 duplicate rows found.

Finding: The sleep table contains 3 duplicate records where the same user has identical sleep data recorded twice for the same night. These duplicates will be removed during the cleaning stage using the DISTINCT function to ensure each night is counted only once per user.

Query 5 : NULL Values Check

Purpose: Identify any missing values in key columns.

null_id	null_date	null_steps	null_calories	null_sedentary
0	0	0	0	0

Finding: No NULL values found in any key columns of the daily activity table. The dataset is complete and requires no imputation or handling of missing values for these fields.

3.2 Cleaning Queries

Query 1 : Create clean daily activity table

Purpose: Remove 61 zero activity days where the device was not worn, standardize the date format, and add day of week and activity category columns to prepare the table for analysis.

Row	total_rows	total_users	earliest_date	latest_date
1	396	34	2016-03-12	2016-04-12

Finding: Clean table successfully created. 61 zero activity days were removed bringing the total from 457 rows down to 396 rows. One user was also removed as all their records were zero activity days, reducing the user count from 35 to 34. Day of week and activity category columns were successfully added and will be used for trend analysis in Phase 4.

Cleaning Query 2 : Sleep Clean Table

Purpose: Remove 3 duplicate sleep records using DISTINCT, convert SleepDay from text format to proper datetime, and add hours asleep, hours in bed, and wasted bed time columns to prepare the table for sleep analysis in Phase 4.

Row	total_rows	total_users
1	410	24

Finding: Clean sleep table successfully created with 410 records from 24 users. 3 duplicate rows were removed using DISTINCT. New columns hours asleep and hours in bed were added by dividing total minutes by 60, making sleep duration analysis straightforward in Phase 4.

Cleaning Query 3 : Hourly Steps Clean Table

Purpose: Convert ActivityHour column from text format to proper datetime and extract the hour of day as a separate column to enable peak activity time analysis in Phase 4.

Row	total_rows	total_users
1	24084	34

Finding: Clean hourly steps table successfully created with 24,084 records from 34 users. ActivityHour column successfully converted from text to datetime format. Hour of day column extracted and will be used to identify peak activity windows throughout the day in Phase 4.

Cleaning Query 4 : Activity and Sleep Merged Table

Purpose: Join the clean daily activity table with the clean sleep table to enable combined analysis of activity and sleep patterns for the same user on the same day.

Row	total_rows	total_users
1	11	11

Finding: The merged table contains 11 records from 11 users. The low overlap occurs because very few users consistently tracked both activity and sleep data on the same dates. This reveals an important behavioral pattern. Users were inconsistent about wearing their device for all tracking features simultaneously. This finding directly supports the recommendation for Bellabeat to encourage more consistent and complete device usage through in-app reminders and engagement features.

3.3 Cleaning Decisions & Rationale

Every cleaning step is documented below. Transparency in data manipulation is a core professional practice.

Cleaning Step	Details & Rationale
Removed zero-activity rows	61 rows removed from dailyActivity where TotalSteps = 0 AND Calories = 0. These represent device non-wear days and would artificially deflate averages.
Removed duplicate sleep rows	3 exact duplicate rows found and removed from sleepDay_merged. Used DISTINCT (SQL) / distinct() (R) on Id + SleepDay.
Parsed date formats	ActivityDate was already in DATE format so no conversion was needed. SleepDay and ActivityHour were stored as STRING and converted using PARSE_DATETIME() in SQL due to AM/PM format.
Added derived columns	day_of_week, hour_of_day, hours_asleep, hours_in_bed, wasted_min, activity_category, total_active_min added for deeper analysis.
Activity classification	Users categorized by avg daily steps: Sedentary (<5,000), Lightly Active (5,000 to 7,499), Fairly Active (7,500 to 9,999), Very Active (10,000 and above).
NULL handling	Fat column in weightLogInfo had NULLs, retained but excluded from analysis. No NULLs found in key dailyActivity columns.

3.4 Dataset After Cleaning

34	396	24	11
Total Users	Cleaned Records	Sleep Users	Weight Users
dailyActivity	After removing 61 zero days	Duplicates removed: 3	32% of users

Phase 4 : ANALYZE

KEY FINDINGS & VISUALIZATIONS

4.1 Summary Statistics

7,555	2,284	7.0h	16h/day
Avg Daily Steps	Avg Calories/Day	Avg Sleep	Avg Sedentary
Median: 6,896	Burned	Rec: 7–9 hours	942 min

4.2 Daily Step Distribution

Purpose: Understand how daily step counts are distributed across all users and identify what proportion fall below the WHO goal.

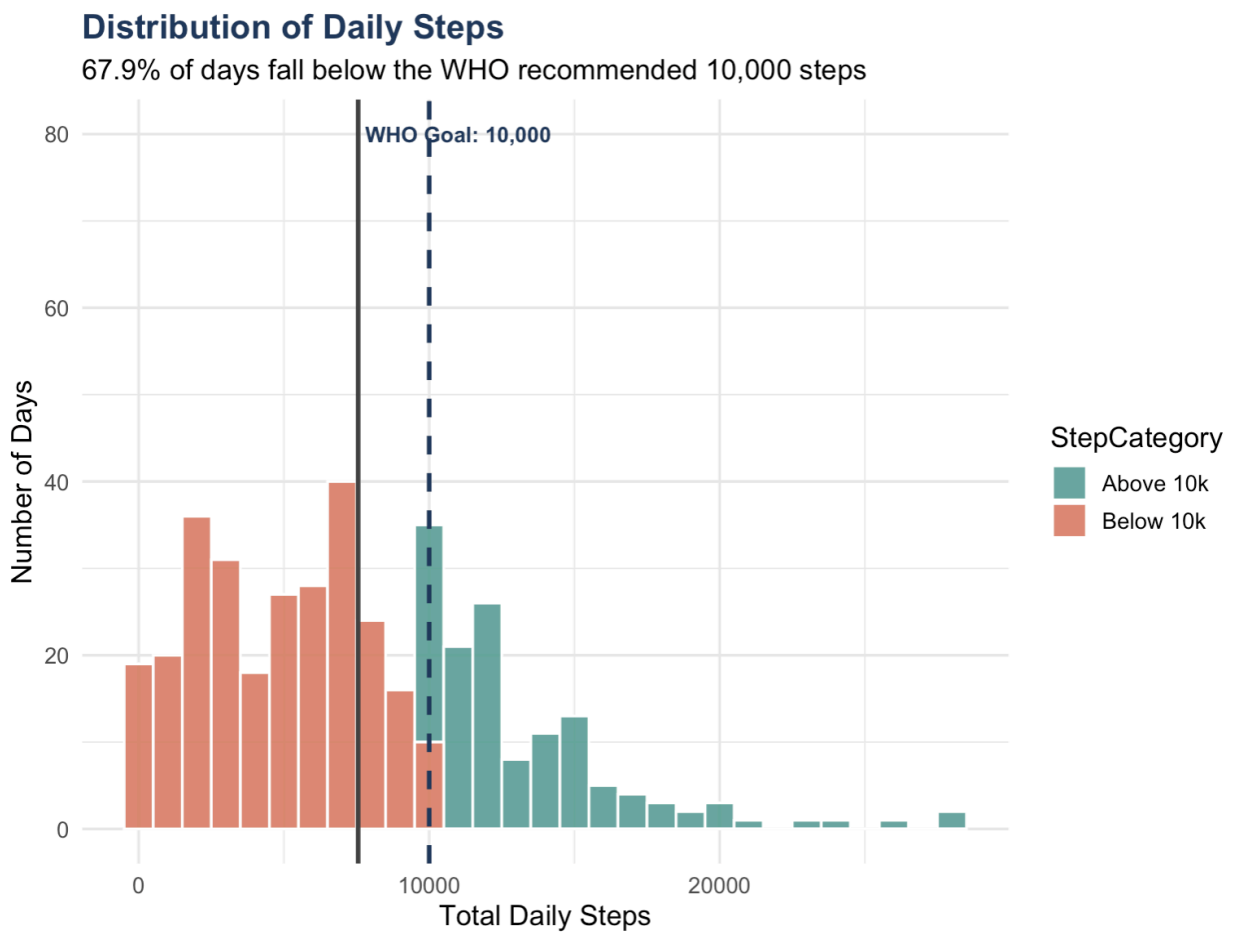


Figure 1: Distribution of Daily Steps, 67.9% of days below the WHO 10,000 step goal

Finding: The distribution reveals that 67.9% of recorded days fall below the WHO recommended 10,000 daily steps. The average of 7,555 steps and median of 6,896 steps both sit well below this target. The gap between the mean and median indicates that a small group of very active users pulls the average upward, meaning the typical user is even less active than the average suggests.

4.3 Activity by Day of Week

Purpose: Identify which days of the week users are most and least active to understand weekly activity patterns.

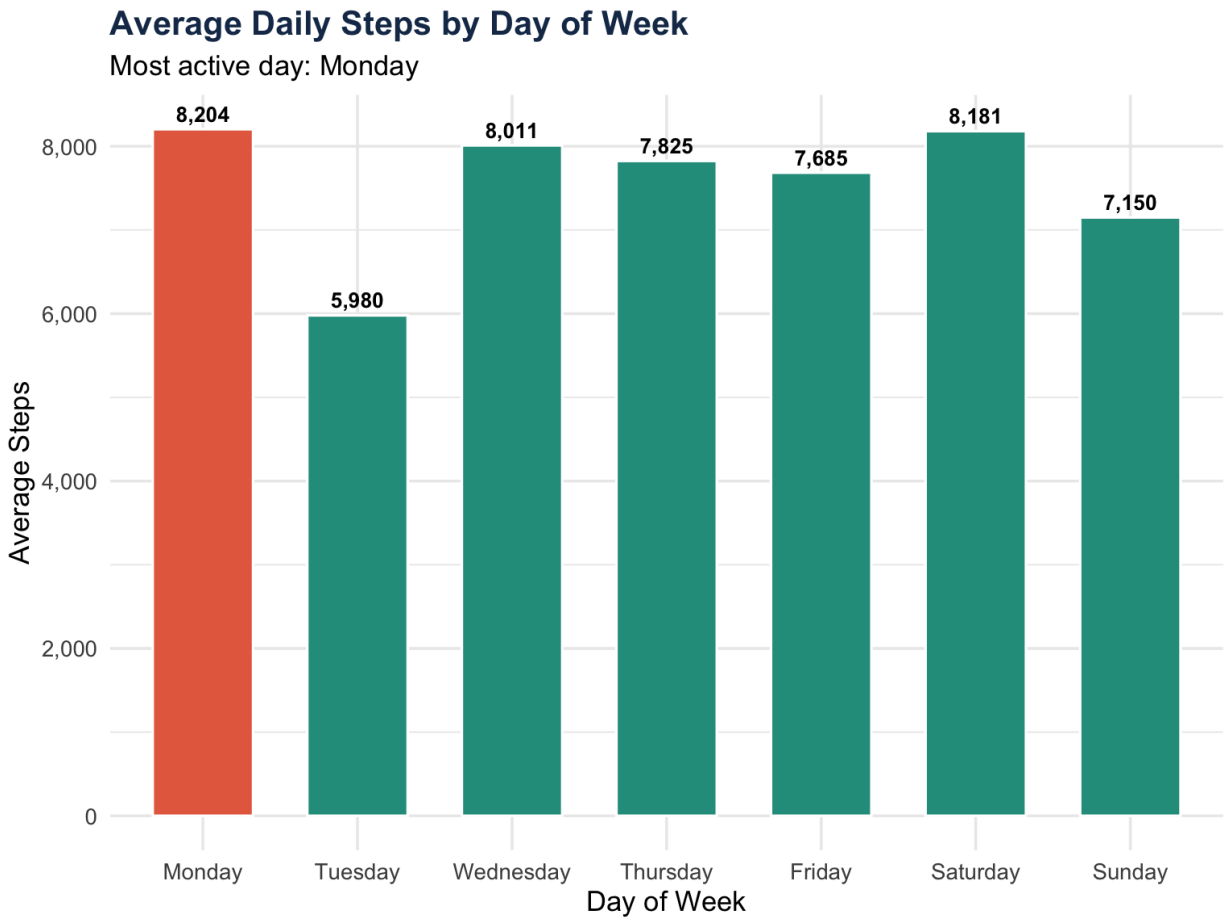


Figure 2: Average Steps by Day of Week, Monday peaks, Tuesday dips

Finding: Monday is the most active day with an average of 8,204 steps while Tuesday records the lowest activity at an average of 5,980 steps, a difference of 2,225 steps. Weekend activity is uneven with Saturday showing stronger engagement than Sunday. This data reveals a consistent weekly rhythm in user activity levels that varies significantly by day.

4.4 Peak Hourly Activity Windows

Purpose: Identify the specific hours of day when users are most physically active to understand daily activity timing patterns.

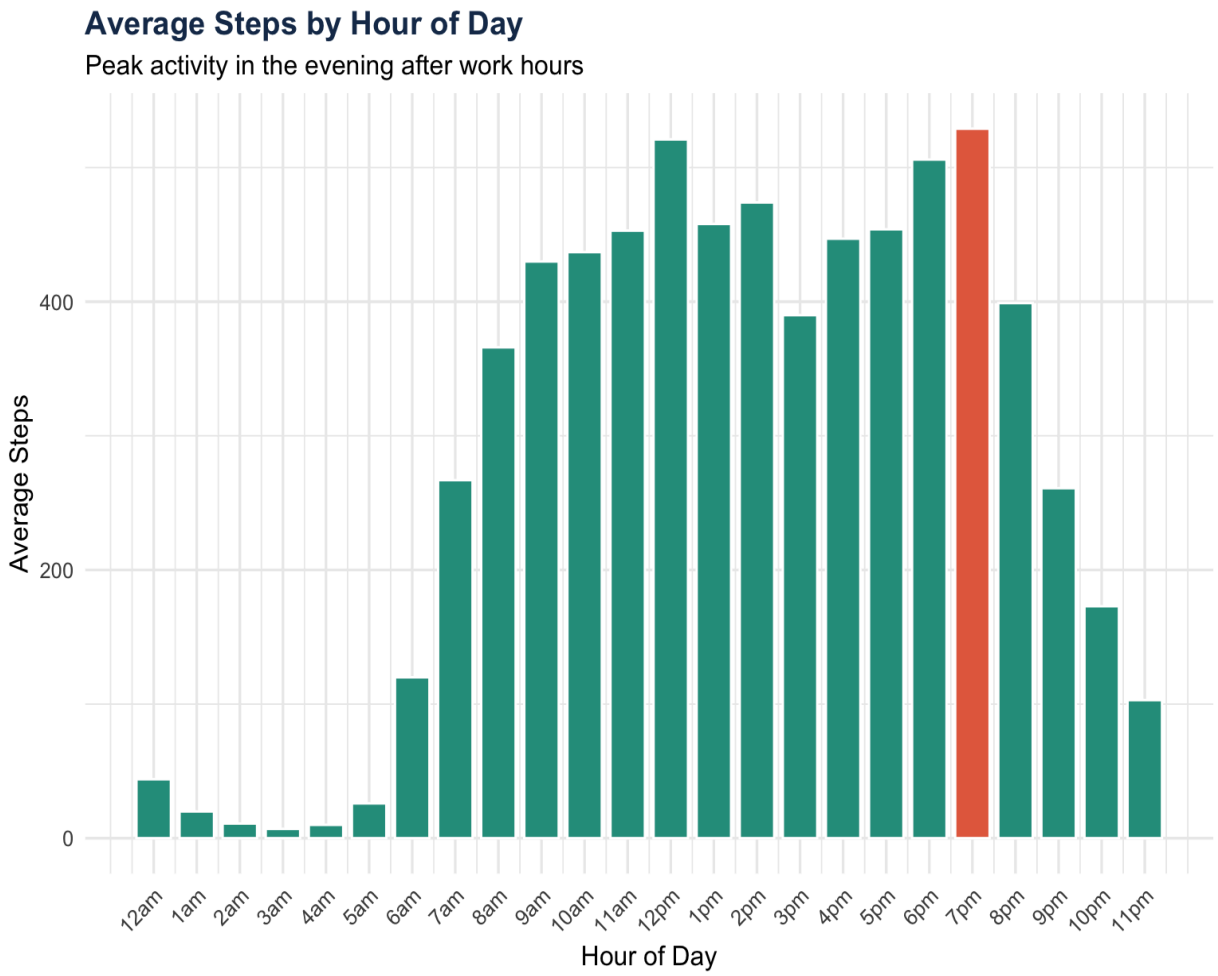


Figure 3: Average Steps by Hour of Day, Peak at evening (6 to 7pm) with lunch secondary peak

Finding: Hourly step data from 34 users reveals two distinct activity peaks throughout the day. The dominant spike occurs in the early evening between 5pm and 7pm corresponding to post work hours. A secondary peak appears around 12pm during lunch hours. Activity drops to near zero between midnight and 5am. Outside of these two windows activity levels remain consistently low throughout the rest of the day.

4.5 Steps vs Calories Correlation

Purpose: Determine whether daily step count has a meaningful relationship with calories burned to validate step goals as a health metric.

Figure 4: Steps vs Calories Burned, Positive correlation (r = 0.56, p less than 0.001)

Finding: There is a statistically significant positive correlation between daily steps and calories burned ($r = 0.56$, $p < 0.001$). Users who walk more consistently tend to burn more calories although the relationship is moderate rather than perfectly linear. This confirms that step count is a meaningful and reliable proxy for caloric expenditure across this user population.

4.6 Sleep Analysis

Purpose: Understand sleep duration patterns across users and identify what proportion fall below CDC recommended minimums.

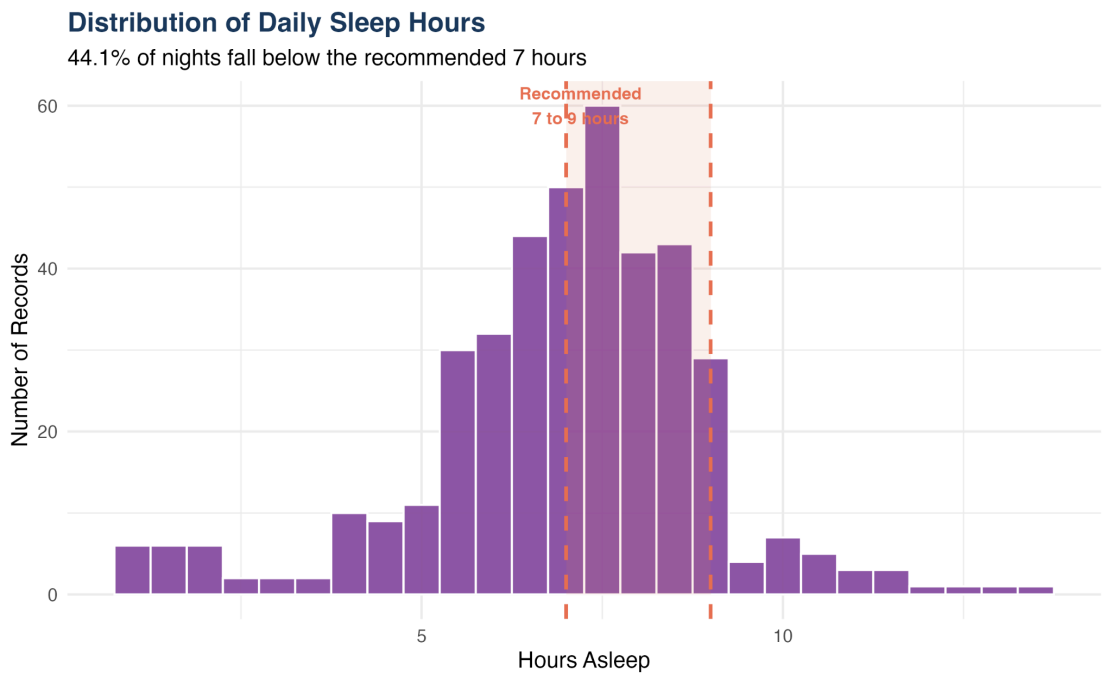


Figure 5: Sleep Distribution, 44.1% of nights below the 7-hour CDC minimum

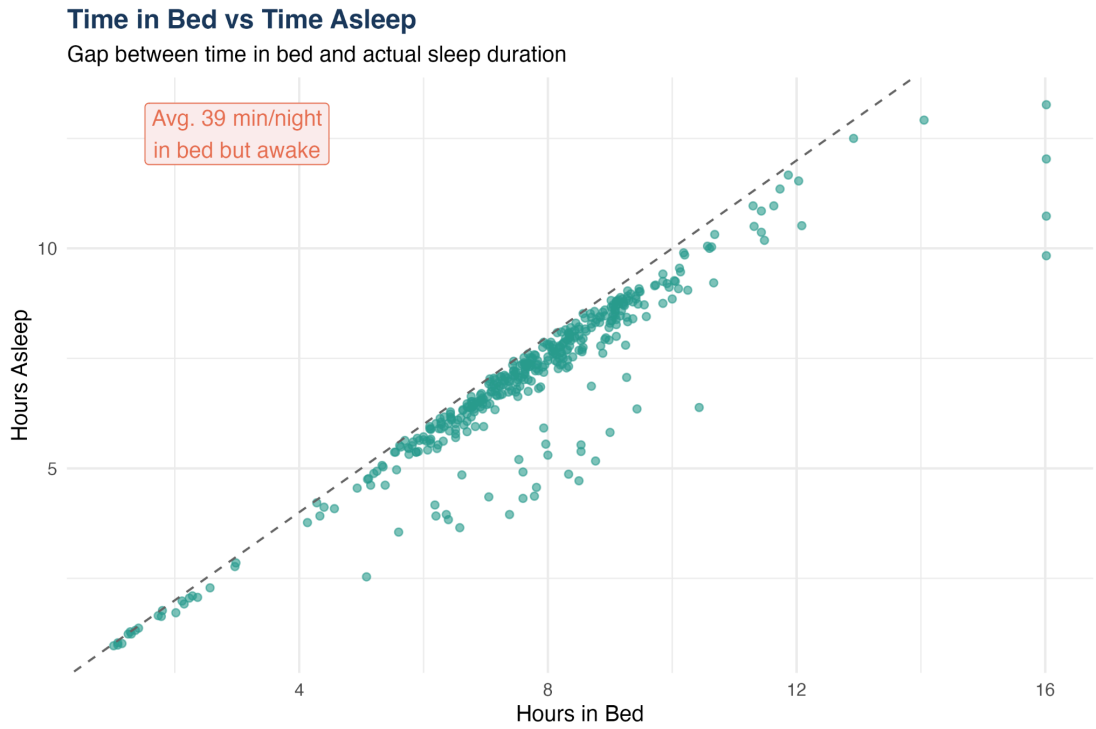


Figure 6: Time in Bed vs Time Asleep, Avg. 39 min/night in bed but awake

Finding: Among the 24 users with sleep data, the average nightly sleep is 6.99 hours, marginally above the CDC minimum of 7 hours. However 44.1% of individual nights still fall below this threshold. Users also spend an average of 39 minutes per night in bed before actually falling asleep, a potential indicator of poor sleep onset quality across the user population.

4.7 User Segmentation

Purpose: Classify users into activity segments based on average daily steps to understand the distribution of activity levels across the user population.

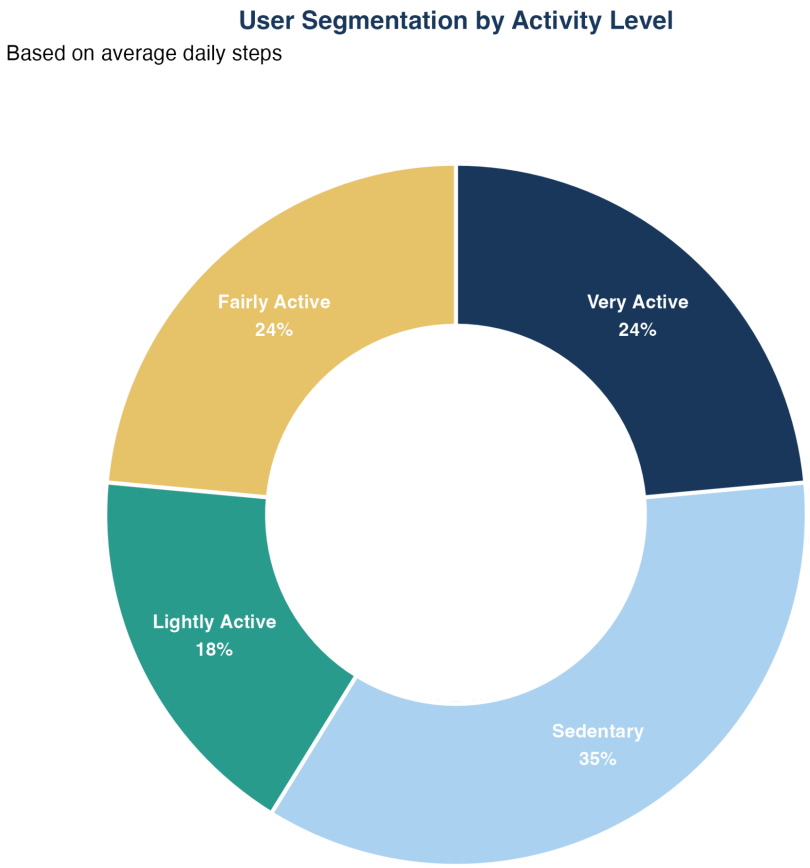


Figure 7: User Segmentation by Activity Level, Bimodal distribution with high sedentary and high very active segments

Segment	Count and Activity Description
Sedentary (<5,000)	12 users (35%), Average steps below 5,000 per day
Lightly Active (5k–7.5k)	6 users (18%), Average steps between 5,000 and 7,500
Fairly Active (7.5k–10k)	8 users (24%), Average steps between 7,500 and 10,000l
Very Active (≥10,000)	8 users (24%), Average steps at or above 10,000

Finding: When users are classified by their average daily steps a bimodal distribution emerges. 12 users (35%) are Sedentary while 8 users (24%) are Very Active. The middle segments of Lightly Active and Fairly Active are smaller, suggesting users in this dataset tend toward the extremes rather than clustering around a moderate activity level.

4.8 Daily Activity Minutes Breakdown

Purpose: Compare the average time users spend in each activity intensity level to understand how the day is distributed between sedentary and active behavior.

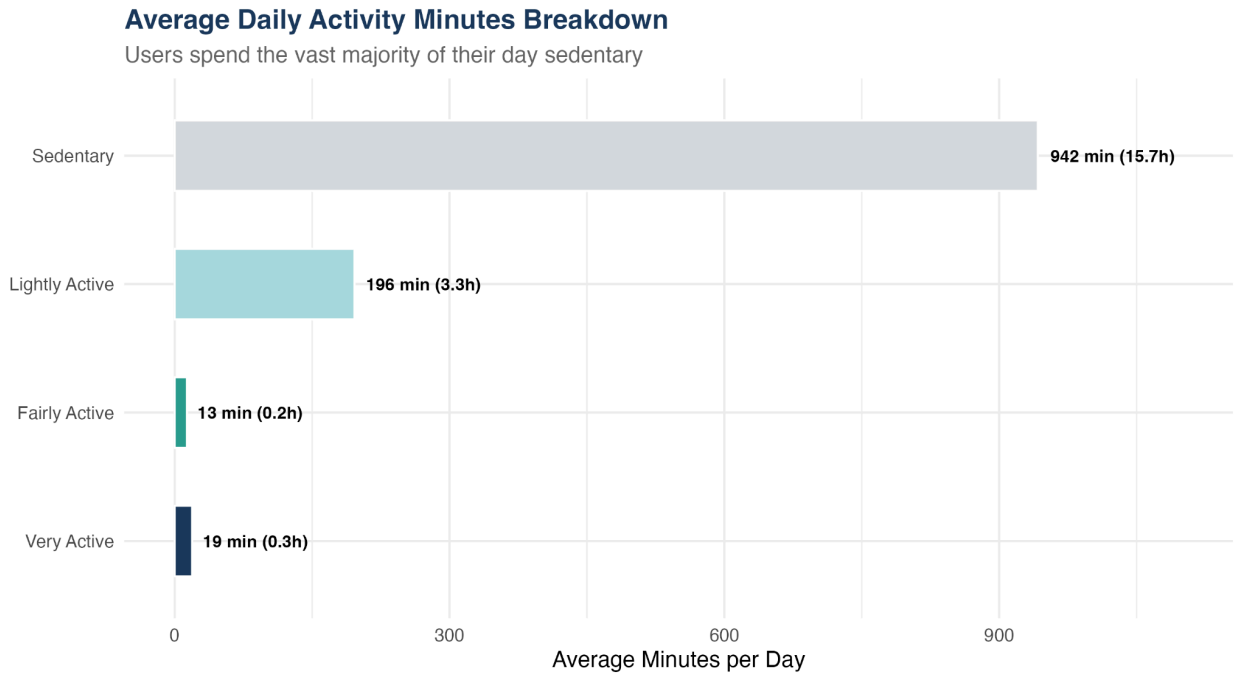


Figure 8: Daily Activity Minutes, Sedentary time (15.7h avg) vastly dominates

Finding: Users average 942 sedentary minutes per day (15.7 hours), compared to just 19 minutes of very active time. Even total active minutes (very + fairly + lightly active combined) fall well short of public health guidelines. The dominance of sedentary time is the defining behavioral pattern of this dataset.

4.9 Weight Tracking Adoption

Purpose: Understand what proportion of users actively logged weight data to assess adoption of manual tracking features.

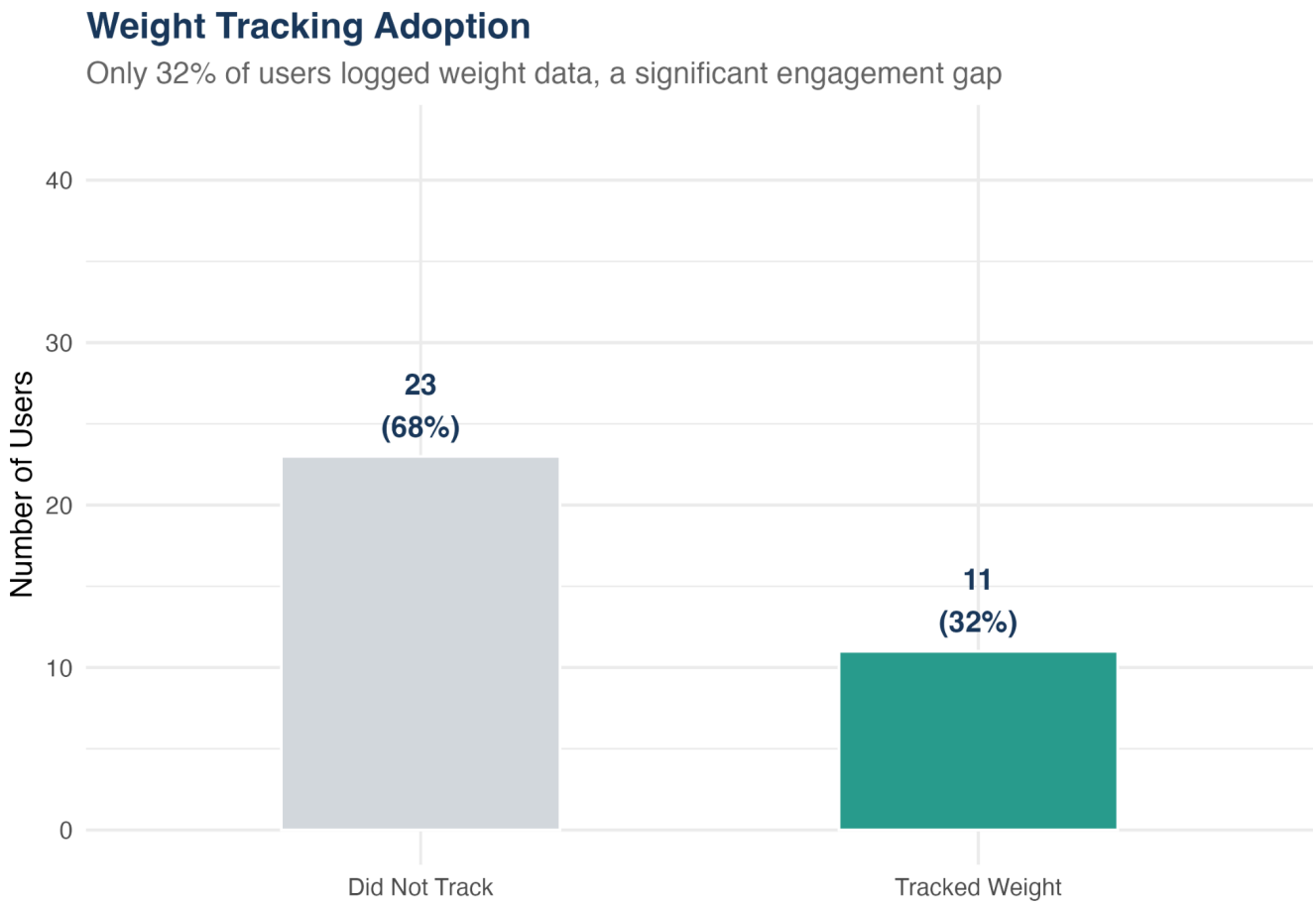


Figure 9: Weight Tracking Adoption, Only 32% of users logged weight data

Finding: Only 11 of 34 users (32%) logged any weight data during the two month study period. The remaining 68% never used this feature at all. This reveals that manual logging features have significantly lower adoption compared to automatic passive tracking features such as steps and calories which recorded data from nearly all users.

Phase 5 : SHARE

CONSOLIDATED FINDINGS SUMMARY

5.1 Key Insights from Analysis

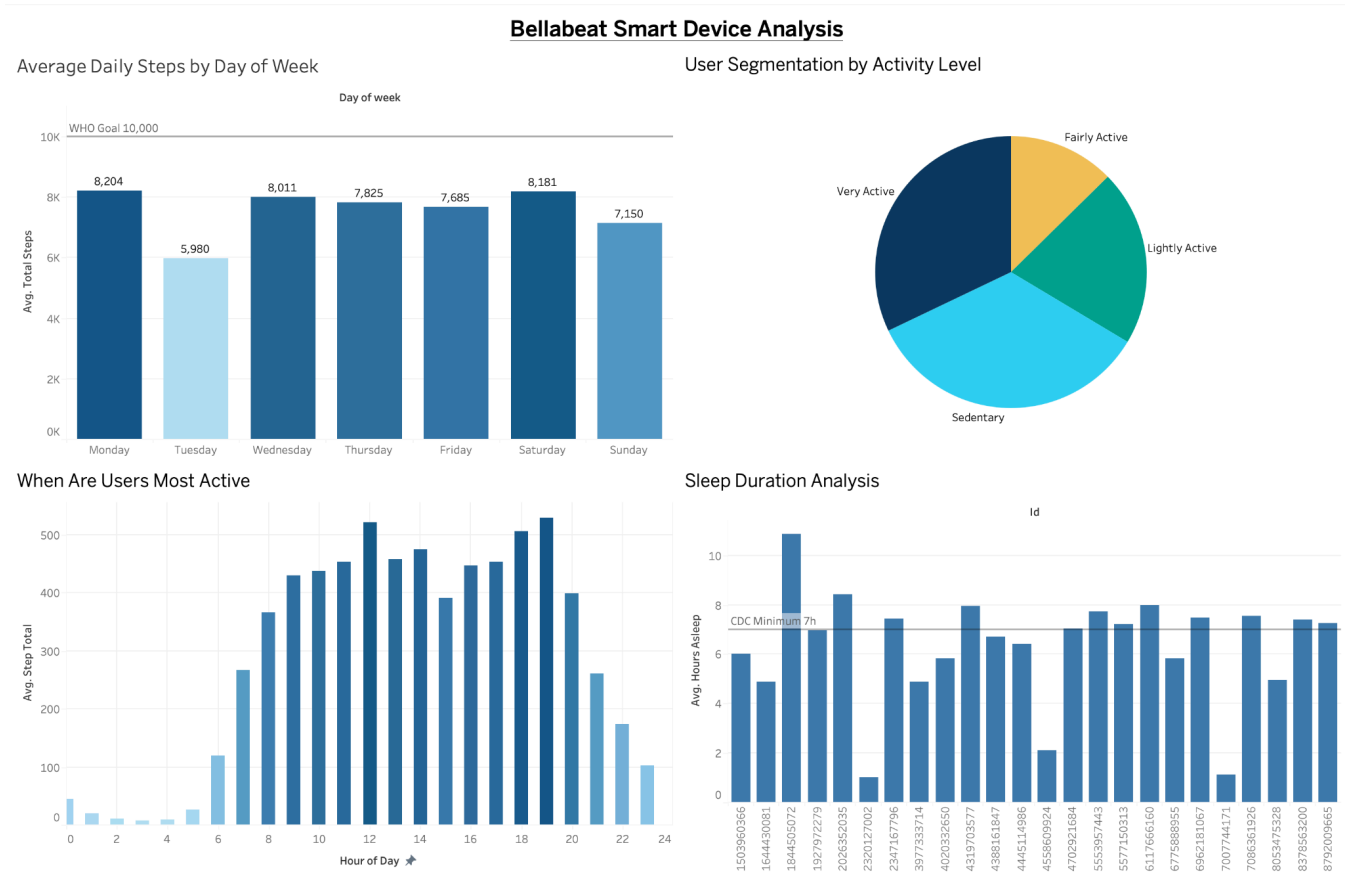
All findings below are drawn directly from the real FitBit dataset collected between March and April 2016. Each finding is supported by specific data points from the analysis.

Finding	Data Evidence
Most users are insufficiently active	Average 7,555 steps per day vs WHO goal of 10,000. 67.9% of days below target
Peak activity is evening & lunch	Strongest hourly spike at 6 to 7pm post work. Secondary peak at 12pm. Minimal activity before 6am.
Monday most active, Tuesday least	2,225-step gap between most and least active days of the week.
Sleep quality needs attention	Average 7.0h sleep. 44.1% of nights under 7 hours. Average 39 min per night in bed awake.
Sedentary time is extreme	Average 15.7 hours sedentary per day vs only 19 minutes very active.
Weight tracking severely underused	Only 32% of users logged weight data. Major feature adoption gap.
Steps and calories are correlated	r = 0.56, p less than 0.001. Step goals are a valid proxy for calorie management.

5.2 Interactive Dashboard

An interactive Tableau Public dashboard was created to present these findings to non-technical stakeholders in a single accessible view. The dashboard combines four visualizations covering daily activity patterns, hourly usage windows, user segmentation, and sleep duration analysis.

Dashboard Link : [Click here to view]



The dashboard contains four views. The top left shows average daily steps by day of week with the WHO 10,000 step reference line. The top right shows user segmentation by activity level across four categories. The bottom left reveals peak hourly activity windows throughout the day. The bottom right shows average sleep duration per user with the CDC recommended 7 hour minimum line.

Phase 6 : ACT

DATA-BACKED MARKETING RECOMMENDATIONS

6.1 Recommendations

Five actionable recommendations for the Bellabeat App marketing team, each grounded in specific findings from the real FitBit data.

Recommendation 1: Gamified Step Challenges

Data: 67.9% of days fall below 10,000 steps. The average user is 2,445 steps short of the WHO goal daily.

Action: Launch gamified daily step challenges inside the Bellabeat App with progress rings, streak rewards, and friend leaderboards. Market this to the Sedentary and Lightly Active segments which together represent 18 of 34 users and have the most room to grow. Campaign message: You are only 2,445 steps from your goal. Let us close the gap together.

Recommendation 2: Smart Notification Timing

Data: Peak activity windows are consistently at 12pm and 6 to 7pm across the user base.

Action: Configure the Bellabeat App to send personalized motivational push notifications at 11:45am and 5:30pm. Market this Smart Timing feature as a key differentiator, positioning Bellabeat not just as a tracker but as an intelligent coach that knows when users are most ready to move. This especially resonates with working women managing structured schedules.

Recommendation 3: Sleep Wind-Down Campaign

Data: 44.1% of nights fall below 7 hours. Users spend an average of 39 minutes in bed awake before falling asleep.

Action: Launch a Sleep Better campaign positioning Bellabeat as a holistic wellness partner. Introduce a sleep wind down mode in the app that activates 30 minutes before a user's target bedtime with guided breathing and phone dimming. Market to users who track activity consistently but neglect sleep, as the data shows these are often the same people.

Recommendation 4: Reduce Sedentary Time

Data: Users average 15.7 hours of sedentary time per day, vastly exceeding active minutes.

Action: Introduce hourly movement reminders in the Bellabeat App when the device detects 60 or more consecutive sedentary minutes. Marketing campaign message: Sitting is the new smoking. Bellabeat nudges you to move every hour. This addresses the root cause of the step deficit more directly than a daily step goal alone. Reducing sedentary time is a bigger opportunity than increasing workout intensity.

Recommendation 5: Simplify Weight Tracking

Data: Only 32% of users tracked their weight. The lowest adoption feature in the dataset.

Action: Reduce friction to a single tap interaction and explore smart scale integration for automatic syncing. Reframe the feature in marketing from a diet tool to a wellness trend view. Campaign message: See how your activity and sleep habits are shaping your body, week over week. Addressing the emotional barrier is as important as fixing the user experience.

6.2 Limitations

This analysis has five key limitations that should be considered before implementing any recommendations.

First, the sample size of 33 to 35 users is too small to draw statistically significant conclusions about the broader population of smart device users or Bellabeat customers. Second, the data was collected in 2016 meaning consumer behavior, device capabilities, and health standards may have changed significantly. Third, the dataset contains no demographic information including age, gender, or location which is a significant gap given that Bellabeat specifically targets women. Fourth, the low overlap between activity and sleep tracking data produced only 11 combined records limiting cross feature analysis. Fifth, no direct Bellabeat product data was available meaning all insights are inferred from FitBit user behavior and must be thoughtfully adapted rather than applied directly.

6.2 Next Steps

To validate and strengthen these recommendations the following actions are proposed.

Collect first party data directly from Bellabeat App users with proper consent and demographic tagging. Run A/B tests on notification timing comparing 12pm versus 6pm delivery to measure which drives more engagement. Partner with sleep researchers to validate the sleep onset intervention before launching the Sleep Better campaign. Repeat this analysis with a larger and more recent dataset of 1,000 or more users from 2023 to 2024. Measure user retention rates by activity segment to quantify the business value of moving users from Sedentary to Lightly Active.

Appendix

TOOLS, CODE & RESOURCES

A.1 Tools Used

Tool	Purpose
SQL (BigQuery)	Data exploration, cleaning, deduplication, table creation, and joins
R with tidyverse and ggplot2	Statistical analysis and data visualization
Tableau Public	Interactive dashboard creation and public sharing